

Notes: Dell & Olken (RES, 2020)

The Development Effects of the Extractive Colonial Economy: The Dutch Cultivation System in Java

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1 Big picture

- **Question.** Can an extractive colonial system have positive long-run local development effects if extraction required building manufacturing capacity, transport infrastructure, and local public-good institutions?
- **Historical setting.** The Dutch Cultivation System forced Javanese villages from the early 1830s through the 1870s to produce export crops. The paper focuses on sugar because it required nearby cane cultivation, water-powered factories, transport links, and large labour mobilization.
- **Two treatments.** The paper separates (i) proximity to Dutch government sugar factories, which created processing infrastructure and agglomeration, from (ii) being inside a forced-cultivation catchment area, which reorganized village land and labour.
- **Main factory result.** Areas near historical sugar factories are more industrialized today, have better historical and modern infrastructure, higher education, and higher household consumption than nearby counterfactual locations that were similarly plausible factory sites.
- **Main forced-cultivation result.** Villages forced to grow sugar cane have more village-owned land, more schooling, higher education, less agriculture, and more manufacturing/retail employment, both historically and today.
- **Mechanism.** Persistence appears to operate through multiple reinforcing channels: local manufacturing linkages, transport infrastructure, agglomeration, village-owned land, and human capital accumulation.
- **Identification.** The factory design compares actual factories to 1,000 simulated counterfactual factory configurations that satisfy suitability, spacing, and geographic balance conditions. The village design uses a spatial RD at the boundaries of sugar catchment areas.
- **Central interpretation.** The results do not imply the Cultivation System was good for people coerced during the nineteenth century. They show that extractive systems can sometimes leave behind productive economic structures that later populations use.
- **Takeaway.** The long-run effect of colonial extraction depends on how extraction was organized. Extraction through forced labour alone may depress long-run development, but extraction that requires local processing and infrastructure can create persistent productive capacity.

2 Historical background and economic mechanism

2.1 The Dutch Cultivation System

- The Cultivation System, or *Cultuurstelsel*, was imposed by the Dutch colonial state in Java beginning in the 1830s.
- It forced villages to produce export crops for the Dutch government. Sugar and coffee dominated profits, but the paper focuses on sugar because sugar production can be localized precisely.
- Sugar was central because raw cane is bulky and perishable. It needed to be processed quickly near the field, which required factories in the countryside.
- The Dutch established 94 water-powered government sugar factories and required nearby villages to supply cane and labour.
- Villages in each factory's catchment area were forced to reorganize land and labour to serve the factory.

Interpretation: The sugar system was extractive, but it was not simply a tax or tribute system. It changed local production technology, transport routes, administrative power, land allocation, and labour organization.

2.2 Why this setting can produce long-run persistence

- The sugar factories introduced manufacturing activity to a rural economy specialized in rice production.
- Cane processing created input–output linkages: upstream machinery and farm inputs, downstream food processing, transport, storage, and local trade.
- The factories required transport infrastructure to bring sugar to ports and global markets.
- Forced-cultivation villages accumulated village-owned land that could later support public goods, especially schools.
- Human capital and infrastructure then reinforce each other: more educated workers make non-agricultural production more feasible, and better roads/rails support markets and specialization.

Interpretation: The paper is not estimating the effect of colonialism as a whole. It estimates how two specific organizational features of colonial extraction changed local development paths.

3 Data and measurement

3.1 Historical Cultivation System records

- The key historical source is the 1858 Umbgrove Commission records held in Dutch archives.
- These handwritten records identify villages that contributed to each sugar factory, the land and labour they provided, and qualitative information about the catchment areas.
- The authors match historical village names to modern geo-referenced locations using the U.S. National Geospatial Intelligence Agency’s Geonames data and historical district information.
- They match 6,383 historical villages to coordinates, corresponding to 2,519 modern village polygons.
- Factories are matched using a similar name/location procedure.

Interpretation: This historical matching is crucial because the spatial RD depends on knowing which villages were actually forced to grow cane and where catchment boundaries were.

3.2 Factory locations and counterfactual factories

- The factory analysis focuses on colonial sugar factories, especially locations without a modern sugar factory nearby, so effects do not simply reflect current sugar processing.
- Counterfactual factory locations are generated using historical and geographic constraints that plausibly shaped Dutch siting decisions.
- Plausible sites must have sufficient nearby sugar-suitable land, be near rivers for water power, satisfy spacing constraints relative to other factories, and maintain aggregate geographic balance.
- The authors use a simulated annealing algorithm to construct 1,000 counterfactual spatial configurations.
- For each outcome, they estimate the same distance-to-factory regression for actual factories and for each counterfactual set.

Interpretation: The counterfactual design asks: relative to other nearby places that could plausibly have hosted a factory, what happened near the places where Dutch factories were actually established?

3.3 Modern and historical outcomes

- **Industrial structure:** SUSENAS household surveys from 2001–2011 measure agricultural, manufacturing, and retail employment. The 1980 Population Census provides earlier employment structure.
- **Economic Census:** The 2006 Economic Census measures manufacturing employment and the value of sugar-related processing.
- **Village infrastructure:** PODES village censuses measure village characteristics such as roads, schools, electricity, and village-owned land.
- **Historical infrastructure:** detailed maps from the Dutch Topographic Bureau measure colonial road and rail density in 1900.
- **Education:** the 2000 Population Census provides years of schooling and school completion by birth cohort, allowing the authors to observe effects for cohorts educated during Dutch rule and later Indonesian periods.
- **Household consumption:** SUSENAS household surveys measure equivalent consumption.

Interpretation: The outcome data span multiple periods and domains. This is important because the paper’s persistence argument is stronger when industrial structure, infrastructure, schooling, and consumption all point in the same direction.

3.4 Key measurement choices

- Distance-to-factory effects are measured using 1-km bins from the nearest historical factory, with the 19–20 km bin omitted.
- Factory proximity regressions include controls for elevation, slope, distance to the coast, rivers, flow accumulation, 1830 residency capitals, the Great Post Road, and VOC ports.
- RD regressions limit the baseline sample to villages within 10 km of the catchment boundary and to land suitable for sugar cultivation.
- Boundary segments are used to ensure comparisons are geographically local.
- Standard errors in RD specifications are clustered by sub-district.

Interpretation: These choices are designed to make comparisons local: villages near the same factory or boundary, with similar geographic fundamentals, are compared to one another.

4 Empirical specifications and interpretation

4.1 Factory-proximity specification: counterfactual factory design

The paper estimates the following equation for actual factories and for each of the 1,000 counterfactual factory configurations:

$$out_v = \alpha + \sum_{i=1}^{20} \gamma_i dfact_v^i + \beta X_v + \sum_j \delta_j fact_v^j + \varepsilon_v. \quad (1)$$

- out_v is the outcome in village or location v .
- $dfact_v^i$ equals one if v is in the i th 1-km distance bin from the nearest actual or counterfactual factory.

- The omitted category is 19–20 km from the nearest factory.
- X_v includes geographic and pre-period controls.
- $fact_v^j$ are nearest-factory fixed effects, so villages are compared to other villages near the same actual or counterfactual factory.

Interpretation: The estimated effect at distance bin i equals the actual-factory coefficient minus the mean of the corresponding coefficients from the 1,000 counterfactual factory configurations. This difference removes patterns that arise merely because plausible factory sites are near rivers, plains, or historically important places.

4.2 Randomization-inference logic

For each distance bin, the authors compare the actual coefficient with the distribution of 1,000 coefficients estimated around counterfactual factories. The p-value is based on the position of the actual coefficient in the counterfactual distribution.

Interpretation: This is not a classical randomized experiment. It is a design-based falsification exercise: if actual factories produce patterns that counterfactual factories rarely produce, the spatial pattern is unlikely to be driven by generic suitability or geography.

4.3 Spatial RD for forced cultivation

The forced-cultivation analysis estimates:

$$out_v = \alpha + \gamma cultivation_v + f(location_v) + g(dfact_v) + \beta X_v + \sum_i \theta_i seg_{vi} + \varepsilon_v. \quad (2)$$

- $cultivation_v$ equals one if village v was inside a forced sugar-cultivation catchment area.
- $f(location_v)$ is a local linear spatial RD polynomial estimated separately for contiguous catchment areas.
- $g(dfact_v)$ is a spline in distance to the nearest historical factory, separating forced-cultivation effects from simple factory-proximity effects.
- seg_{vi} are boundary segment fixed effects, comparing nearby villages along the same border segment.
- The baseline uses a 10 km bandwidth and a triangular kernel.

Interpretation: This specification isolates the impact of being compelled to grow cane, conditional on being near the same factory system. The RD compares villages just inside the catchment boundary with nearby villages just outside.

4.4 What each design identifies

- **Factory design:** long-run effects of the establishment of sugar processing factories and associated infrastructure/agglomeration.
- **RD design:** long-run effects of village-level forced cane cultivation and the organizational changes imposed on subjected villages.
- **Combined interpretation:** the Dutch system created two complementary persistent changes: production centres and reorganized villages.

Interpretation: The two empirical strategies are complementary. The first asks what happened near factories. The second asks what happened to villages forced into the cane-growing catchments.

5 Identification strategy and threats

5.1 Factory location: why counterfactuals are credible

- Factories had to be near sugar-suitable plains and rivers because cane was heavy and water power was required.
- Historical accounts indicate that Dutch factory-location knowledge in the 1830s was coarse; detailed agronomic and population surveys were unavailable.
- The authors construct counterfactual factories that satisfy the same broad constraints, so the design does not compare factory areas to arbitrary non-sugar areas.
- Geographic balance checks show no consistent relationship between proximity to actual factories and elevation, slope, distance to rivers, coast, pre-period roads, residency capitals, or VOC ports.

Interpretation: The key identifying assumption is that actual factory proximity is not systematically correlated with unobserved local potential after conditioning on broad suitability, geography, and counterfactual factory patterns.

5.2 Counterfactual contamination and spillovers

- If counterfactual factories are too close to actual factories, actual effects could contaminate the counterfactual distribution.
- The authors argue that effects decline quickly enough that this bias is limited and would tend to flatten estimates toward zero.
- The shape of the estimated effects matters: strong impacts near the actual factory and flat effects further away are more consistent with local treatment effects than with broad regional trends.

Interpretation: Counterfactual contamination is a conservative concern. If actual factory effects spill into counterfactual comparisons, the estimated actual-minus-counterfactual effects become smaller.

5.3 RD identification for forced cultivation

- The RD assumes potential outcomes vary smoothly across catchment boundaries in the absence of forced cultivation.
- Boundary segment fixed effects make the comparison local along the same boundary.
- The specification controls flexibly for factory proximity, so the coefficient is not simply capturing being closer to a factory.
- Table 1 tests balance in pre-treatment geography and historical access variables.
- The authors discuss migration/sorting: forced cultivation was historical, and today's population density is not higher in treated villages, making mass selective persistence less likely.

Interpretation: The RD design is stronger than a simple treated-control comparison because the treatment changes sharply at a boundary, while many geographic variables change smoothly.

5.4 Main threats and paper responses

- **Pre-existing productivity:** addressed through geographic controls, counterfactual factories, and RD balance tests.
- **Modern sugar processing:** addressed by focusing on historical factories without modern factories nearby.
- **General regional development:** addressed by nearest-factory fixed effects, boundary segment fixed effects, and counterfactual configurations.
- **Spatial spillovers:** the authors emphasize that effects are concentrated near factories, not across the whole surrounding region.
- **Sorting or migration:** the RD evidence and population-density comparisons reduce, but cannot completely eliminate, this concern.

Interpretation: The identification case is cumulative. No single test proves exogeneity, but the combination of historical institutional detail, counterfactual simulations, boundary discontinuities, and robustness checks makes the interpretation plausible.

6 Tables and figures

6.1 Figure 1: Counterfactual factories example and Figure 2: Counterfactual factories heatmap

Read:

- Figure 1 illustrates the construction of counterfactual factory locations: actual factory, a plausible counterfactual point satisfying suitability constraints, and a set of alternative counterfactual sites.
- Figure 2 shows the density of counterfactual factories across Java; plausible counterfactuals cluster around sugar-suitable regions and near actual factory areas.
- Black dots are government sugar factories; grey areas are sugar-suitable; darker shading indicates more counterfactual factory placements.

Interpretation: These figures explain the identifying counterfactual for the factory analysis. The paper does not compare factory locations to all of Java; it compares them to other places that could plausibly have hosted factories given geography and technology.

Economic message: The counterfactual design is the core empirical innovation for estimating long-run effects of sugar processing infrastructure.

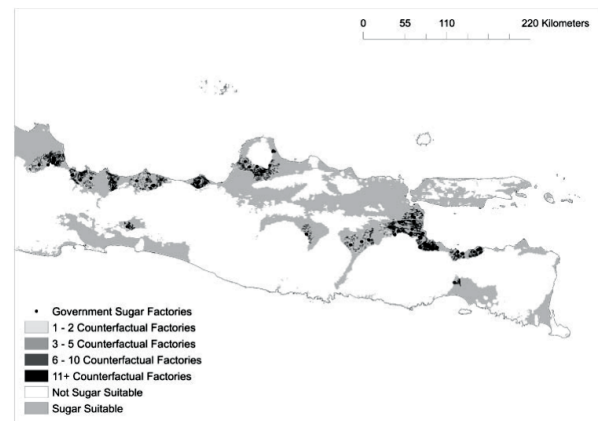
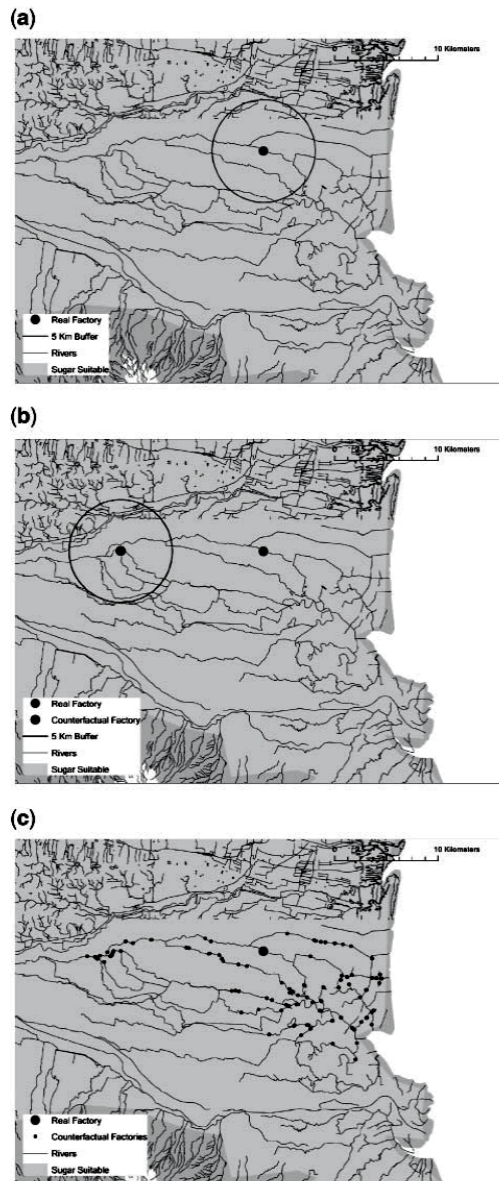


FIGURE 2
Counterfactual factories heatmap.

Black dots are the government sugar factories in the main sample. Gray areas have an elevation and slope that are suitable for :
cultivation whereas white areas do not. Within areas suitable some shading indicates the density of the counterfactual factories.

6.2 Figure 3: The Cultivation System and Figure 4: Construction of catchment areas

Read:

- Figure 3 maps government sugar factories and forced-cultivation catchment boundaries across Java.
- Figure 4 shows how catchment areas are constructed from factory locations, matched points, historical boundaries, and modern village polygons.
- The maps visually separate two treatments: factory proximity and being inside a forced-cultivation catchment.

Interpretation: These figures show why the paper uses two empirical strategies. Factories generate distance-based exposure, while catchments generate sharp RD treatment boundaries.

Economic message: The Cultivation System created both industrial nodes and administrative/cultivation territories. Both can leave persistent effects.

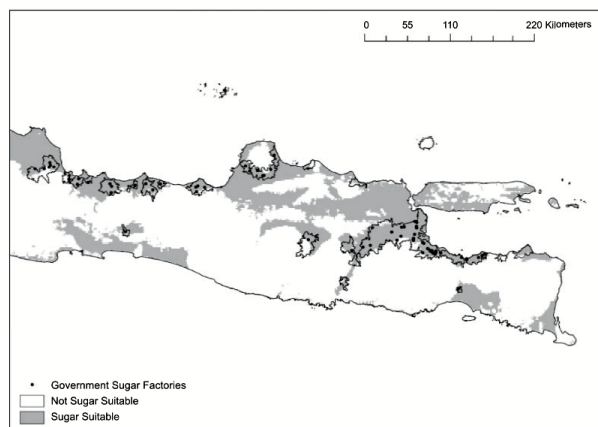


FIGURE 3
The Cultivation System.

Sources: [Commissie Umbgrove \(1858\)](#). Black dots are government sugar factories. Lines are forced cultivation catchment area boundaries. Gray areas have an elevation and slope that are suitable for sugar cultivation, whereas white areas do not.

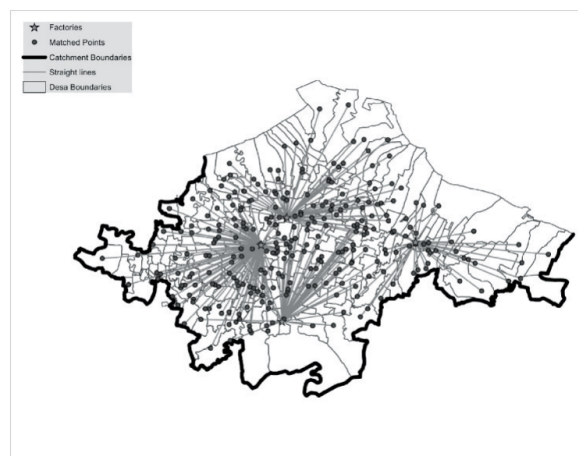


FIGURE 4

6.3 Figure 5: Geography

Read:

- Figure 5 plots distance-to-factory coefficients for pre-period or geographic characteristics: elevation, slope, distance to coast, flow accumulation, distance to river, distance to 1830 residency capital, distance to the Great Post Road, and distance to VOC port.
- Most series do not show strong, systematic gradients with factory proximity.
- The magnitudes are generally small relative to the means, supporting the plausibility of balance.

Interpretation: This is a factory-location balance figure. If actual factory locations were systematically placed at better locations in ways not captured by controls or counterfactual factories, these geographic variables would likely show strong patterns.

Economic message: The factory results are less likely to be driven by obvious geographic fundamentals.

6.4 Figure 6: Share in agriculture: illustration of methodology

Read:

- Figure 6 illustrates the methodology using the share of workers in agriculture.
- Counterfactual distributions are shown for both independent and common shifts.
- The actual coefficients show a strong negative effect near factories: villages close to factories are less agricultural.
- The counterfactual coefficients are generally much smaller, showing that the actual gradient is unusual relative to plausible placebo factory placements.

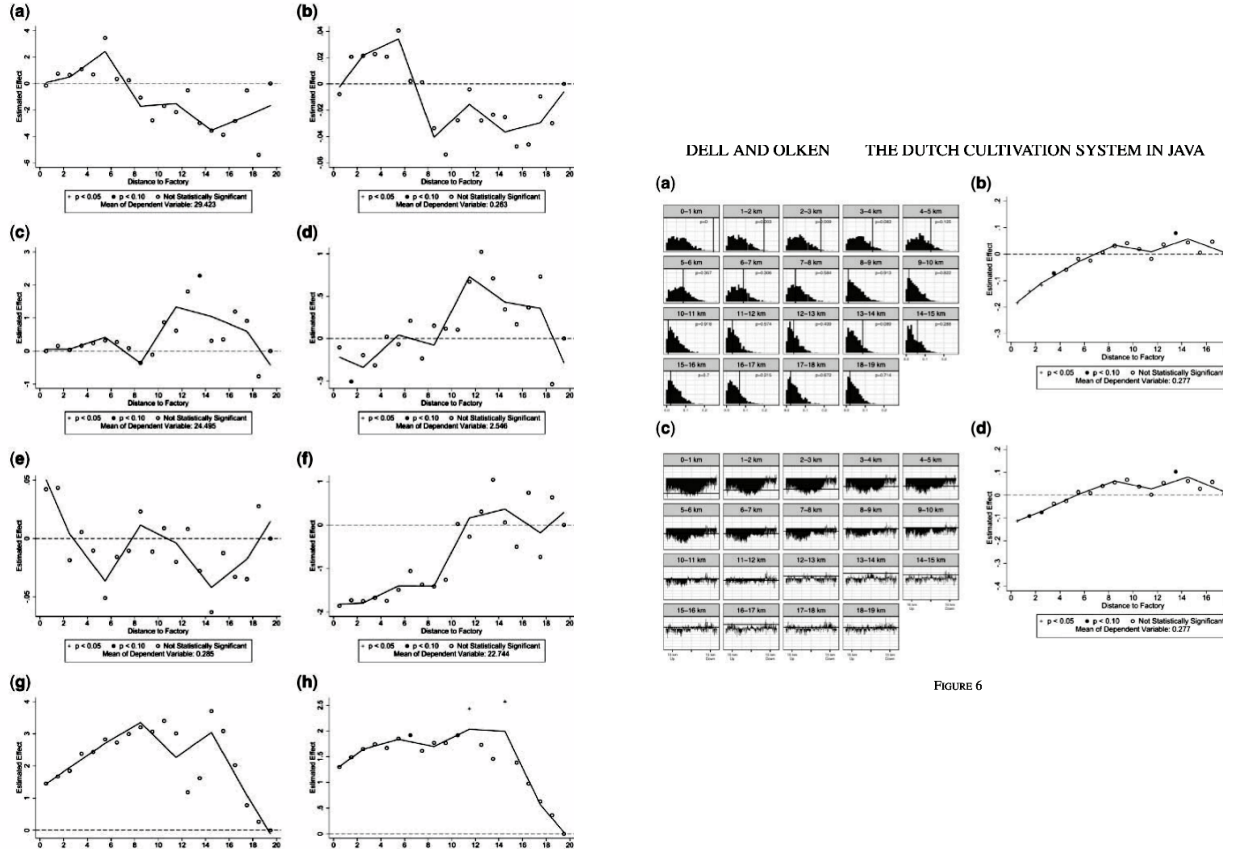


FIGURE 6

Interpretation: The figure demonstrates how actual-minus-counterfactual effects are constructed and how p-values are derived from counterfactual distributions.

Economic message: Proximity to sugar factories shifted local employment away from agriculture, a core sign of persistent structural transformation.

6.5 Figure 7: Industry and agglomeration

Read:

- Figure 7 shows strong negative effects on agricultural employment near historical factories and positive effects on manufacturing and retail.
- The same qualitative pattern appears in recent SUSENAS data and in the 1980 census.
- Population density and log population are higher near factories, consistent with agglomeration.

Interpretation: The pattern indicates more than the persistence of sugar itself. It suggests a broader local structural change toward non-agricultural activity and higher density.

Economic message: Historical factory locations became persistent local economic centres.

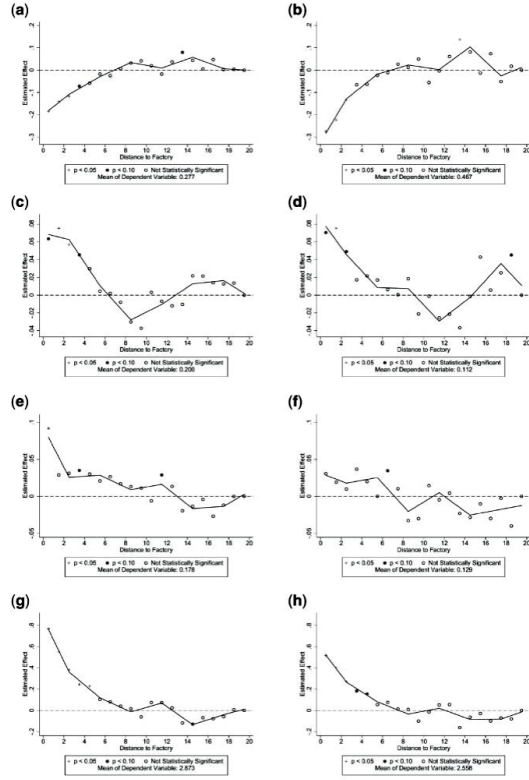


FIGURE 7

Industry and agglomeration. (a) Agriculture (Susenas 2001–11), (b) Agriculture (Census 1980), (c) Manufacture (Susenas 2001–11), (d) Manufacturing (Census 1980), (e) Retail (Susenas 2001–11), (f) Retail (Census 1980), (g) Pop. Density (PODES 2003), (h) Log Pop. Density (PODES 1980).

Notes: These figures plot coefficients estimated from regressing the outcome variable on 1-km bins of distance to the nearest factory, controlling for nearest-factory fixed effects and geographic and pre-period characteristics. The means of analogous esti

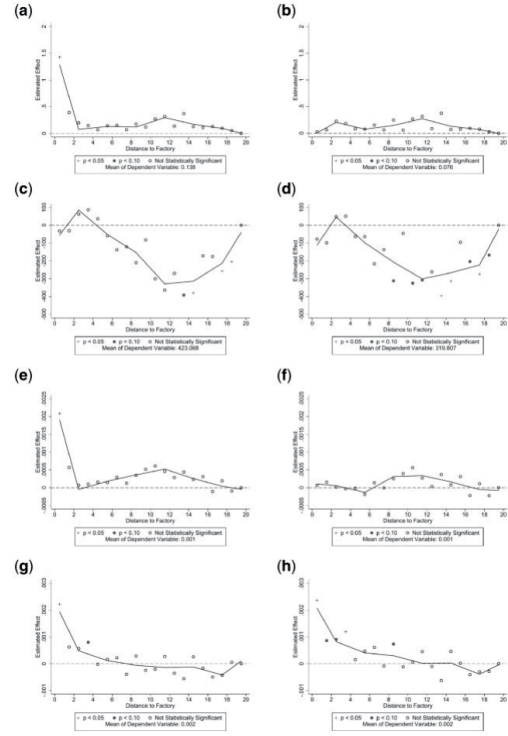


FIGURE 8

Sugar and linked industries. (a) Log Value Sugar Processed (Full Sample, Economic Census 2006), (b) Log Value Processed (No Modern Factories, Econ Census 2006), (c) Tons of Cane Grown (Full Sample, PODES 2003), (d) To Cane Grown (No Modern Factories, PODES 2003), (e) Employment Share Upstream (Full Sample, Economic Ce 2006), (f) Emp Share Upstream (No Modern Factories, Economic Census 2006), (g) Employment Share Down-str (Full Sample, Economic Census 2006), (h) Emp Share Downstream (No Modern Factories, Economic Census 2006). Notes: These figures plot coefficients estimated from regressing the outcome variable on 1-km bins of distance to the nearest factory, controlling for nearest-factory fixed effects and geographic and pre-period characteristics. The means of analogous esti

6.6 Figure 8: Sugar and linked industries

Read:

- Figure 8 separates direct sugar outcomes from input–output linked industries.
- Historical factory proximity predicts value of sugar processed and tons of cane grown in the full sample.
- Even when modern sugar factories are excluded, downstream employment remains higher near historical factories.
- Upstream effects are weaker; downstream food-processing-related effects are more pronounced.

Interpretation: This figure supports the input–output linkage mechanism. The persistence is not only because sugar processing survived; linked downstream industries also concentrate near historical factory sites.

Economic message: Industrial persistence is transmitted through sectoral linkages and local agglomeration after the original colonial industry disappears.

6.7 Figure 9: Infrastructure

Read:

- Figure 9 shows that areas near historical factories had greater colonial road and railroad density in 1900.
- Modern infrastructure effects also persist: local and intercity roads and railroads remain denser near factories.
- Some historical infrastructure effects are very concentrated within a few kilometers of factories.

Interpretation: The figure supports infrastructure as a channel of persistence. Factories required transport to ports, and that infrastructure appears to have shaped later economic geography.

Economic message: Extractive infrastructure can have persistent development effects when it remains useful for later commerce.

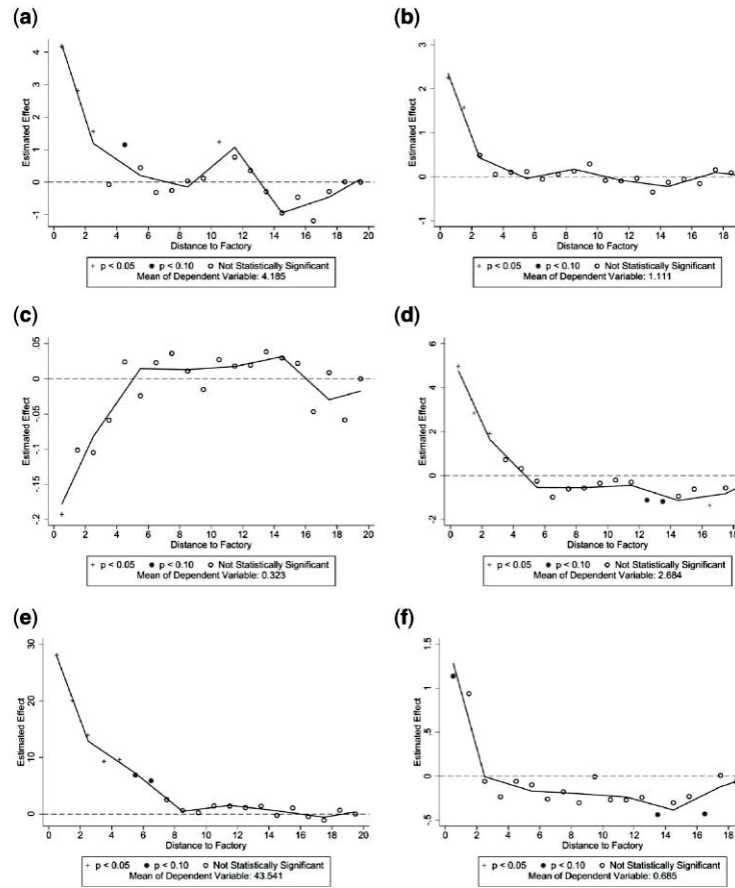


FIGURE 9

Infrastructure. (a) Colonial Road Density (1900), (b) Colonial Railroad Density (1900), (c) Dirt Road (PODES 19 (d) Intercity Road Density, (2017) (e) Local Road Density (2017), (f) Railroad Density (2017).

Notes: These figures plot coefficients estimated from regressing the outcome variable on 1-km bins of distance to the nearest hist factory, controlling for nearest-factory fixed effects and geographic and pre-period characteristics. The means of analogous esti

6.8 Figure 10: Other public goods and Figure 12: Expenditure

Read:

- Figure 10 shows positive effects near factories on electricity and high schools, especially historically rare high schools in 1980.
- The distance-to-subdistrict-capital effect is negative near factories, consistent with more local administrative or economic centrality.
- Figure 12 shows household expenditure is higher near historical factories, around 10% in the paper's discussion for villages immediately adjacent to factories.

Interpretation: The public-goods patterns show that factory regions did not merely specialize in industry; they also accumulated complementary infrastructure and human-capital institutions.

Economic message: Better public goods and higher consumption indicate that the factory effects translated into broader welfare-relevant outcomes.

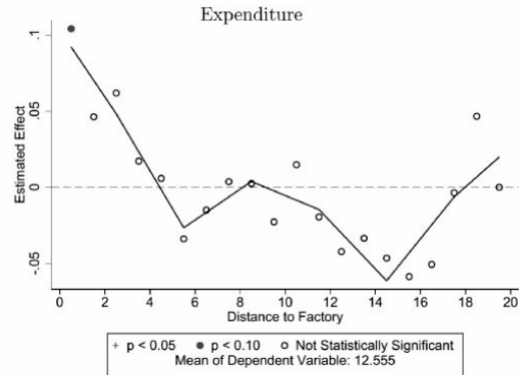
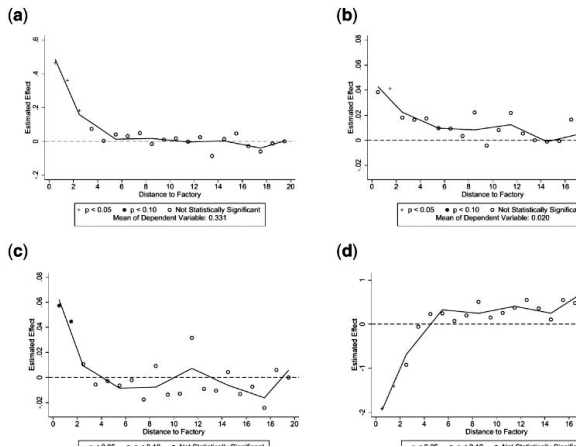


FIGURE 12

Expenditure.

Figure 12 plots coefficients estimated from regressing the outcome variable on 1-km bins of distance to rolling for demographic variables, survey year fixed effects, nearest-factory fixed effects, and geographics. The means of analogous estimates computed from 1,000 counterfactual factory configurations are the points are fit with a linear spline. p -values compare the effect of proximity to the nearest actual factory to the nearest counterfactual factory, computed from 1,000 counterfactual factory configurations.

6.9 Figure 11: Education and Figure 13: Education by cohort

Read:

- Figure 11 shows positive effects near factories on years of schooling and completion outcomes.
- Cohort panels indicate that education effects appear for cohorts educated under Dutch rule and for cohorts educated after independence.
- Figure 13 focuses on subjected villages and shows positive effects across cohorts, with primary completion effects especially strong for cohorts born during primary-school expansion.

Interpretation: The education results support a human-capital channel of persistence. Historical organization and public goods continued to influence outcomes over multiple generations.

Economic message: Human capital accumulation is a bridge from colonial production structures to modern development.

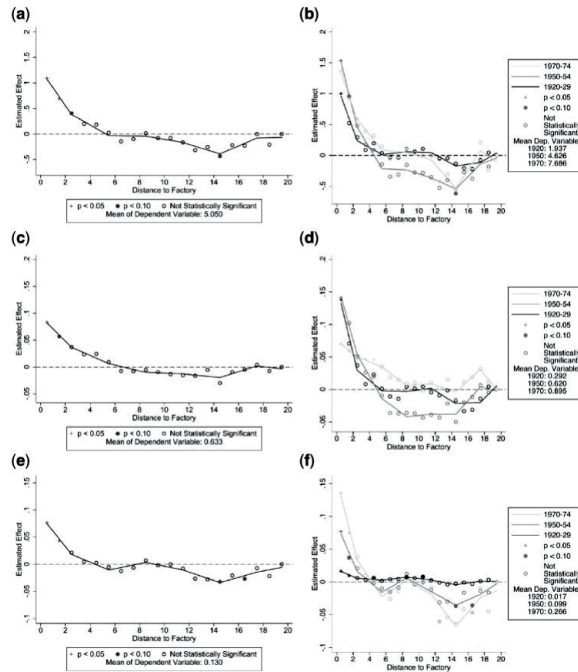


FIGURE 11

Education. (a) Years Education (2000 Census), (b) Years Education by Cohort, (c) Primary (2000 Census), (d) Primary by Cohort, (e) High School (2000 Census), (f) High School by Cohort.

Notes: These figures plot coefficients estimated from regressing the outcome variable on 1-km bins of distance to the nearest historical factory, controlling for gender, nearest-factory fixed effects, and geographic and pre-period characteristics. The means of analysis

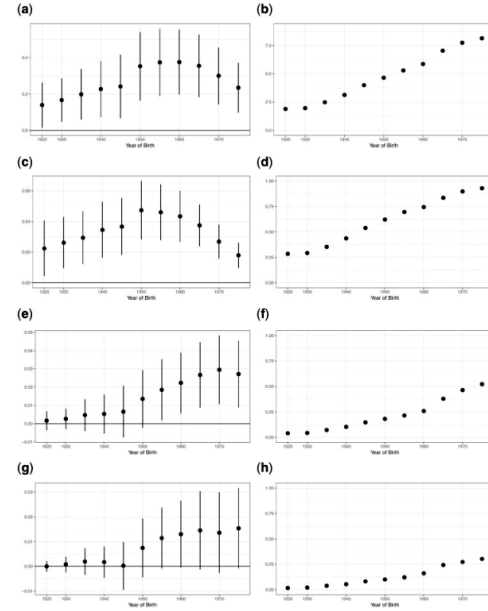


FIGURE 13

Education by cohort: subjected villages (2000 census). (a) Years of schooling, (b) Years of Schooling Levels, (c) Primary Completion, (d) Primary Completion Levels, (e) Junior High Completion, (f) Junior High Completion Levels, (g) High School Completion, (h) High School Completion Levels.

Notes: In the left panels, each point plots a separate regression coefficient for different birth cohorts (1920-9, 1930-4, 1935-9, ..., 1975-9). Lines show 90% confidence intervals. In the right panels, points plot means. The unit of analysis is the individual, and the specification includes gender dummies, geographic and pre-period characteristics, boundary segment fixed effects, a linear spline in distance to the nearest historical factory, and a linear polynomial in latitude and longitude estimated separately for each catchment area. Robust standard errors are clustered by sub-district.

6.10 Table 1: Geographic characteristics: subjected villages

Read:

- Table 1 tests balance for villages inside versus outside forced-cultivation boundaries.
- Elevation, slope, distance to coast, flow accumulation, distance to river, distance to residency capital, and distance to the Great Post Road are largely balanced.
- The distance to VOC port differs modestly, but the overall balance pattern supports the RD design.

Interpretation: This table is the main RD balance check. The treatment boundary should not coincide with large pre-existing geographic differences.

Economic message: The forced-cultivation estimates are more credible if villages just inside and outside the boundary are geographically similar.

TABLE 1
Geographic characteristics: subjected villages

	Elevation	Slope	Distance to coast	Log flow accumulation	Distance to river	Distance to 1830 residency capital	Distance to Great Post Road	Distance to VOC Port
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cultivation	-1.908 (0.744)	-0.017 (0.013)	0.117 (0.106)	-0.041 (0.160)	0.001 (0.012)	0.062 (0.133)	-0.060 (0.118)	0.265 (0.136)
Obs	4,553	4,553	4,553	4,549	4,553	4,553	4,553	4,553
Clusters	383	383	383	383	383	383	383	383
Mean	31.17	0.26	24.90	2.56	0.29	24.80	25.16	30.23

Notes: The unit of observation is the village. Regressions include boundary segment fixed effects, a spline in distance to the nearest historical factory with kinks each 3 km, and a linear polynomial in latitude and longitude estimated separately for each catchment area. Robust standard errors, clustered by sub-district, are in parentheses.

Land tenure: subjected villages

	Village land 2003	Land share	Village land 1980	Land share	99th Percentile	90th Percentile	50th Percentile	10th Percentile
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cultivation	2,339 (0.851)	0.015 (0.004)	3,631 (1,559)	0.013 (0.005)	0.032 (0.150)	0.860 (1.130)	0.215 (0.191)	-0.046 (0.141)
Obs	4,550	4,550	4,205	4,107	4,202	4,202	4,202	4,202
Clusters	383	383	380	380	381	381	381	381
Mean	18.61	0.09	23.95	0.11	3.60	17.00	3.78	4.27

Notes: The unit of observation is the village. Regressions include geographic and pre-period characteristics, boundary segment fixed effects, a linear spline in distance to the nearest historical factory, and a linear polynomial in latitude and longitude estimated separately for each catchment area. Robust standard errors, clustered by sub-district, are in parentheses.

6.11 Table 2: Land tenure: subjected villages

Read:

- Subjected villages have substantially more village-owned land in 2003 and 1980.
- In 2003, the point estimate for village land total is 2.339 and the land share effect is 0.015.
- In 1980, the corresponding total is 3.631 and the share effect is 0.013.
- Private agricultural land inequality measures are not strongly affected.

Interpretation: The land results match historical accounts that forced cultivation empowered village officials and increased communal or office-holder controlled land.

Economic message: Village land plausibly became a fiscal base for local public goods after the Cultivation System ended.

6.12 Table 3: Education: subjected villages

Read:

- Subjected villages have higher years of schooling in the 2000 Census, with an effect of 0.220 years relative to a mean of 5.10.
- They have higher primary and junior-high completion; the high-school effect is positive but smaller.
- In 1980, subjected villages have fewer people with no schooling and slightly higher primary completion.
- Village heads are more likely to have completed high school.

Interpretation: The table connects forced cultivation to human capital, not just to land. The result is consistent with village-owned land or local organization increasing school provision.

Economic message: The institutional changes imposed for extraction may have raised the local capacity to finance and value education.

Education: subjected villages								
	Years education (1)	2000 Population census Primary school (2)	Junior high (3)	High school (4)	1980 Census No school (5)	Primary school (6)	Village head Years education (7)	High school (8)
Cultivation	0.220 (0.074)	0.025 (0.006)	0.015 (0.007)	0.007 (0.005)	-0.018 (0.008)	-0.004 (0.010)	0.094 (0.086)	0.028 (0.013)
Obs	16,125,747	16,125,747	16,125,747	16,125,747	653,313	653,188	26,630	26,630
Clusters	383	383	383	383	358	358	383	383
Mean	5.10	0.64	0.27	0.13	0.41	0.19	11.87	0.74

Notes: The unit of observation is the individual. Regressions include boundary segment fixed effects, a linear spline in distance to the nearest historical factory, geographic and pre-period characteristics, and a linear polynomial in latitude and longitude estimated separately for each catchment area. Columns (1) through (6) include gender dummies, and columns (7) and (8) include survey year fixed effects. Robust standard errors, clustered by sub-district, are in parentheses.

Industrial structure: subjected villages							
	SUSENAS (2001–11)			1980 Population Census			Log Equiv. consumption (8)
	Ag. (1)	Manuf. (2)	Retail (3)	Ag. (4)	Manuf. (5)	commerce (6)	Num. Manuf. firms (7)
Cultivation	-0.042 (0.012)	0.031 (0.010)	0.012 (0.007)	-0.036 (0.021)	0.008 (0.013)	0.027 (0.010)	21,734 (8,639)
Obs	130,335	130,335	130,335	127,873	127,873	127,873	4,549
Clusters	381	381	381	358	358	358	383
Mean	0.27	0.21	0.18	0.48	0.11	0.12	71.72

Notes: The unit of observation is the individual in columns (1) through (6), the village in column (7), and the household in column (8). The sample is restricted to men age 18–55 in columns (1) through (6). Regressions include geographic and pre-period characteristics, boundary segment fixed effects, a linear spline in distance to the nearest historical factory, and a linear polynomial in latitude and longitude estimated separately for each catchment area. Columns (1) through (3) include survey year fixed effects. Robust standard errors, clustered by sub-district, are in parentheses.

6.13 Table 4: Industrial structure: subjected villages

Read:

- Subjected villages have less agricultural employment and more manufacturing and retail employment in SUSENAS 2001–2011.
- The agriculture coefficient is -0.042 , manufacturing is 0.031 , and retail is 0.012 .
- Similar patterns appear in the 1980 Census, with less agriculture and more commerce/manufacturing.
- The number of manufacturing firms is higher in subjected villages.

- Log equivalent consumption is not meaningfully higher in the RD table, indicating that forced cultivation effects differ from factory-proximity consumption effects.

Interpretation: Forced-cultivation villages also shifted away from agriculture, but the channel is not simply the same as factory proximity. The mechanism likely involves land, schooling, and local organization.

Economic message: The village-level coercive system left a persistent imprint on occupational structure.

7 Interpretive synthesis

7.1 Why positive long-run effects are possible despite extraction

The paper’s central contribution is to separate contemporaneous welfare from long-run production structure. Forced cultivation likely harmed many people during the colonial period, but the Dutch also had to build a local economic system that could generate a surplus. Once the colonial regime disappeared, some of the production structure remained useful.

- Factories created local manufacturing nodes.
- Roads and railroads connected those nodes to ports and markets.
- Downstream industries agglomerated near old processing centres.
- Forced-cultivation villages acquired land and public-good capacity.
- Education effects persisted across cohorts.

Interpretation: The paper therefore reframes colonial extraction as heterogeneous. The key question is not whether extraction was extractive, but what economic organization extraction required.

7.2 How the two designs fit together

- The factory-counterfactual design explains persistent industrial, infrastructure, and consumption effects near processing centres.
- The spatial RD explains land, schooling, and occupational changes within forced-cultivation villages.
- Together, the results suggest persistence through both physical capital and local institutions.

Interpretation: The factory design is about agglomeration and infrastructure; the RD design is about village organization and human capital. The conclusion is stronger because both designs point toward structural transformation.

7.3 What the paper does not claim

- It does not claim colonialism was beneficial overall.
- It does not estimate the total welfare effect for people alive during the Cultivation System.
- It does not claim all extractive institutions raise development.
- It does not identify what would have happened if Java had industrialized through inclusive institutions instead.

Interpretation: The relevant counterfactual is not a just or inclusive industrialization program; it is whether the specific extractive production structure left persistent local effects relative to plausible nearby alternatives.

8 Short summary

- Dell and Olken study the Dutch Cultivation System in Java, an extractive colonial system centred on sugar production.

- The Dutch established water-powered sugar factories and forced nearby villages to grow cane.
- The paper separates factory-proximity effects from forced-cultivation village effects.
- Factory effects are identified by comparing actual factories to 1,000 plausible counterfactual factory configurations.
- Forced-cultivation effects are identified using a spatial RD at catchment boundaries.
- Areas near factories are less agricultural, more manufacturing/retail oriented, more connected by roads/rails, more educated, and richer.
- Effects persist even where modern sugar factories are absent, implying persistence through broader agglomeration and linked industries.
- Subjected villages have more village-owned land, more schooling, and a less agricultural occupational structure.
- Land and public goods provide a plausible channel from forced cultivation to long-run education and development.
- The paper emphasizes that the long-run effect of colonial extraction depends on the production system created by extraction.

9 Conclusion

9.1 Core conclusion

The Dutch Cultivation System transformed local economic organization in Java. Areas near sugar factories and villages subjected to cane cultivation show persistent differences in modern industrial structure, infrastructure, education, and public goods. These effects are not simply general colonial effects; they arise from the particular way Dutch extraction required local processing, transport, and village reorganization.

9.2 Economic intuition

Extractive production can be locally productive when extraction requires fixed industrial investments and supporting infrastructure. Sugar had to be processed quickly near the field, so the Dutch created local factories and roads. Villages that grew cane acquired institutional and land-tenure changes that could later support public goods. These changes created a persistent development path.

9.3 Takeaway

The paper's lasting insight is that the legacy of colonial extraction is not one-dimensional. Extractive institutions can have negative institutional and welfare consequences while also leaving productive infrastructure and organization. Whether the long-run effect is positive or negative depends on the specific production technology, infrastructure needs, and institutional changes required by the extractive system.